



:

2026

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Lely

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- — , , ML
- : + +

Lely .

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- ① (35+)
- ② API Rust/Axum (25+)
- ③ - SvelteKit (24+)
- ④ ML 11
- ⑤ Lely Horizon API (23)
- ⑥ CI/CD

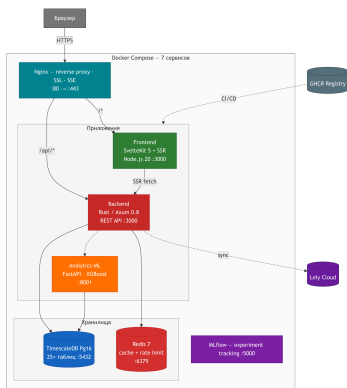
Backend

- Rust + Axum
- SQLx + PostgreSQL
- TimescaleDB
- JWT + bcrypt

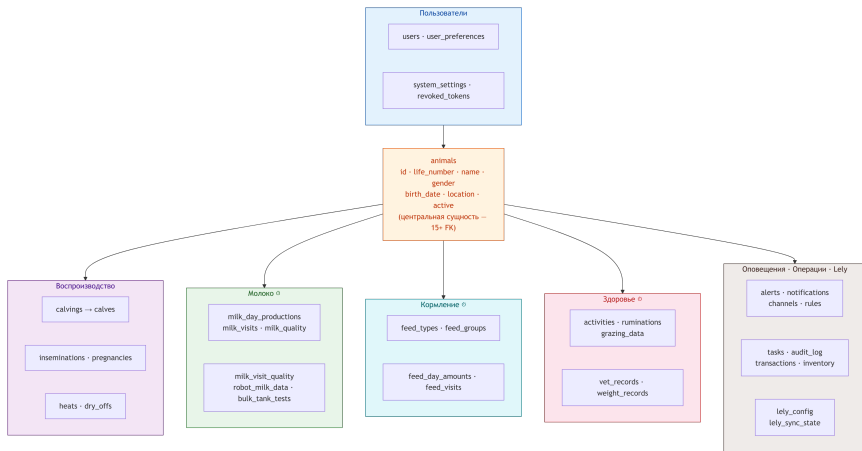
Frontend

- SvelteKit 2
- Svelte 5 (Runes)
- Tailwind CSS 4
- Chart.js

- Docker Compose
- Nginx + SSL
- Prometheus + Grafana
- GitHub Actions
CI/CD

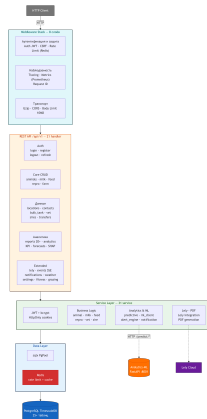


- Nginx — , SSL,
- Backend — -, Lely,
- Analytics-ML — 11 (Python/FastAPI)



- 35+ PostgreSQL TimescaleDB
- 23 , 10

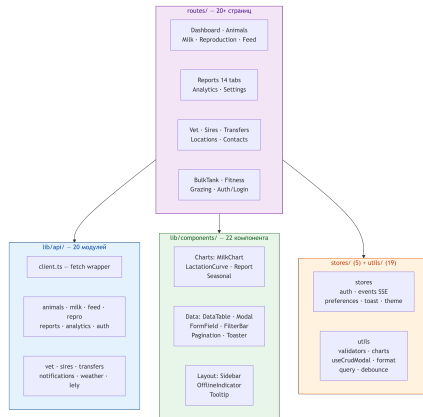
Backend:



Backend:

- JWT (access + refresh), HS256
- HttpOnly, SameSite=Strict cookies
- Token revocation
- : admin / user
- Rate limiting (Redis / in-memory)
- CSRF- (Origin header)
- AES-256-GCM (Lely)
- Prometheus metrics

Frontend:



- 20+ , 20 API-, 22 , 19
- SSR + PWA, Chart.js , Tailwind CSS 4

()	XGBoost	+ q10/q90
()	Prophet	+
	XGBoost	
	XGBoost	
	XGBoost	
	XGBoost	
	KMeans	
	Isolation Forest	

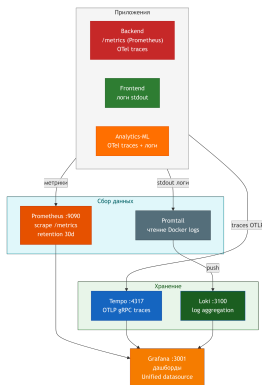
MLOps: , Optuna , , SHAP , MLflow

- — — , (7)
- — - $y(t) = a \cdot t^b \cdot e^{-ct}$, $t_{\max} = -b/c$
- — z^- , SCC,
- —

Lely Horizon

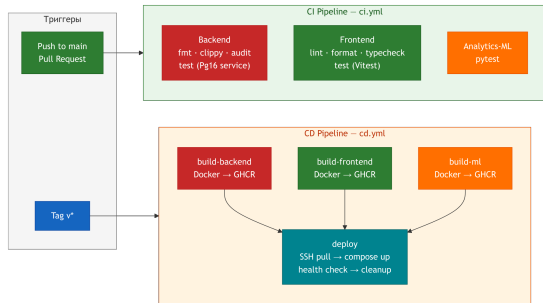
- 5
- 23 : , , , , , , , ,
- (7)
- AES-256-GCM
- 58 , retry
- -

(Grafana LGTM)

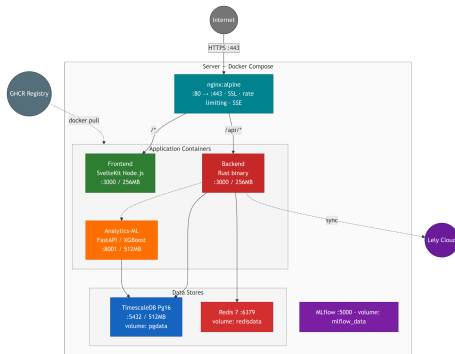


- Prometheus — backend, ML, Redis (30d retention)
- Grafana —
- Loki + Promtail — Docker-
- Tempo — OTLP

CI/CD (GitHub Actions)



(Docker Compose)



- CI: fmt, clippy, audit, test (3)
- CD: Docker build → GHCR → SSH deploy + health check
- Prod: SSL, rate limiting, resource limits, log rotation

- 25+ API
- 33 -
- 35+
- 24+ frontend
- 11
- 17+ (CSV/PDF)
- 30+

- Rust —
- SvelteKit — UI
- XGBoost —
- Docker —
- Grafana LGTM —
- CI/CD —

- :
- Lely (23 ,)
- ML 11 : , , , , , ,
- Docker , SSL CI/CD
- : , ,

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